

Home automation has evolved into the connected home, which is a broad system of home electronics that has so much intelligence it can quickly repay owners for upfront investment costs through efficiency savings and in a relatively short amount of time. The connected home promises to bring further automation into the home, linking together disparate technologies, appliances and devices, and coordinating them into a synergistic platform to serve homeowners, all within the control of their fingertips. As a simple and fully integrated system, home automation offers home occupants peace of mind, which in today's chaotic and turbulent world is of enormous economic value.

One aspect of home automation that helps drive peace of mind involves home security and monitoring. This article articulates how leading-edge innovation and electronic component technologies are supporting advanced smart home security and monitoring solutions. These advances stem from new developments in electronics, technology and design areas involving power, sensing, compute, command and control, and communications. Home automation is conveniently remaking the home environment, making it safer and more secure.

Security and monitoring

Consider for yourself a home automation system that could provide security and monitoring along with a wide variety of entertainment options. This home would incorporate infrastructure that can transmit and receive secure high-speed internal and external communication. It would do so

in a manner that both optimises and minimises home resource consumption of utilities such as power, water, gas and data. This automated home would have the intelligence to allow it to act as its own steward over low-end tasks within the home, operating in a manner analogous to a well-intentioned house sitter. This type of automated house would be both smart and connected.

Today's electronic components have become ever more affordable while simultaneously increasing exponentially in performance capabilities. Today's communication infrastructures and services have become wonderfully standardised, convenient and ubiquitous. Integration of both passive and active semiconductors to produce these complex infrastructures has brought forth a network so powerful that it equates to a personal, 24/7, around-the-clock, electronic butler in your home, on your property.

Automated homes consist of multiple separate systems that one integrates to work together. In its simplest form, one might envision an entire network as connecting in a manner as shown in the illustration in Fig. 1.

Since our focus is on security and monitoring, we will forego further discussion on the three other blue blocks in Fig. 1. Three areas that we will not discuss in this article are:

- Smart heating, ventilation and air-conditioning (HVAC), and smart lighting control
- Smart meters for utility monitoring
- Communication and entertainment

Before the development of the smart home, false alarms of intruder detection could account for up to 98 per cent of security triggers. This high level of false positives is expensive and ends up producing a syndrome analogous to the *Aesop's Fables* classic, *The Boy Who Cried Wolf*.

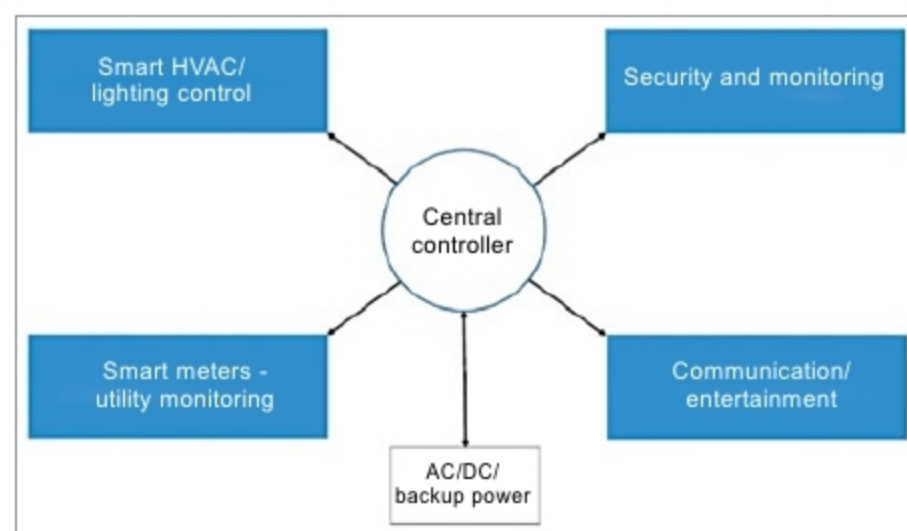


Fig. 1: Connected homes host a network of devices (Credit: AVX)



LOW COST LCD / LED MODULES



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LC135

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